

P230038 ARSAMALI TASKS COAL LAB4

Auxiliary flag:

Code :

```
org 0x0100
```

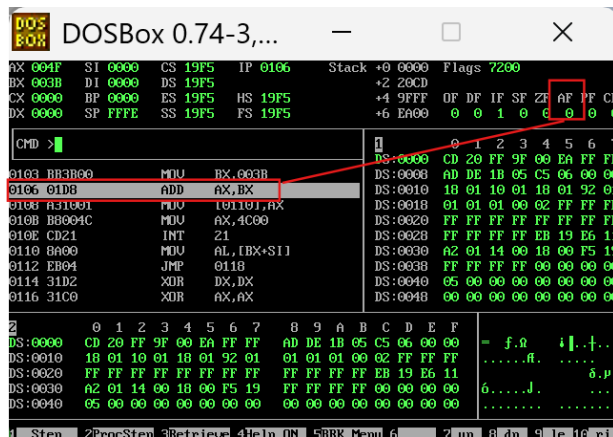
```
mov ax, 01001111b ;
mov bx, 00111011b ;
add ax, bx ;
mov [result], ax
```

```
mov ax, 0x4c00
int 0x21
```

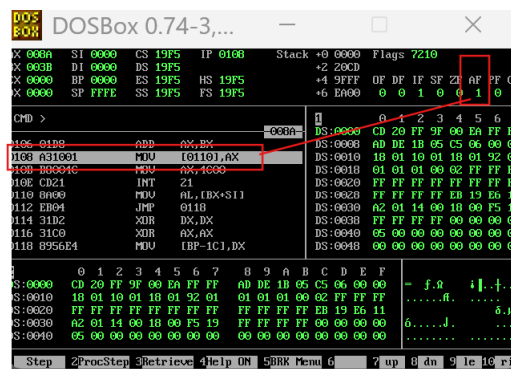
```
result dw 0
```

<-(The squared part will raise auxiliary flag).

As we see below auxiliary flag is zero until Ax and BX are added.



ONCE THE COMMAND IS RUN IT BECOMES 1 (AS CARRY GOES FROM THE FIRST NIBBLE TO THE SECOND):



PARITY FLAG:

```
org 0x0100
```

```
start:
```

```
mov ax, 00000001b
```

```
mov bx, 00000010b
```

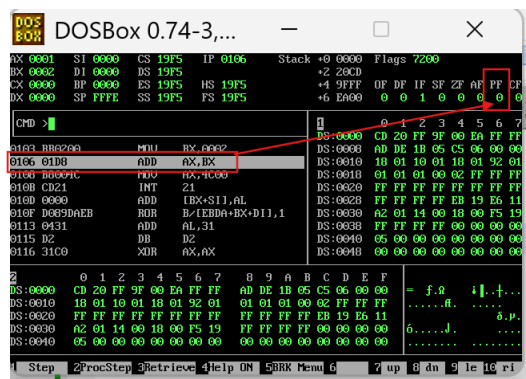
```
add ax, bx
```

```
mov ax, 0x4c00
```

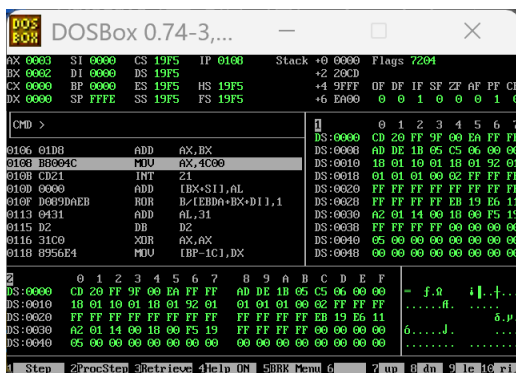
```
int 0x21
```

```
result dw 0
```

<-AX=1(0001), BX=2(0010), AX+BX=3(0011)



AS AX AND BX HAS ODD PARITY (ODD NUMBER OF ONES) PARITY FLAG IS NOT RAISED



AS AX AND BX ARE ADDED IT RESULT IN EVEN PARITY (0011) THIS PARITY FLAG IS RAISED.

CARRY FLAG:

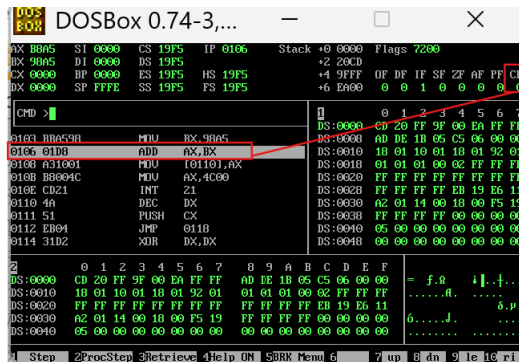
```
org 0x0100

start:
    mov ax, 1011100010100101b
    mov bx, 1001100010100101b
    add ax, bx
    mov [result], ax

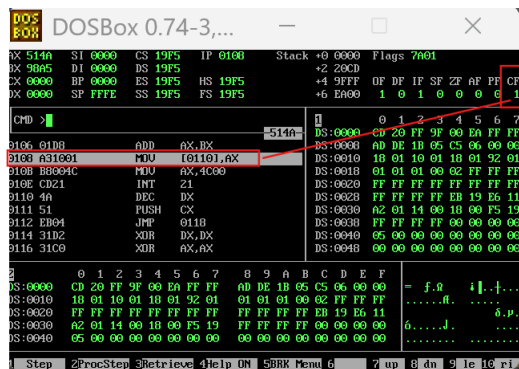
    mov ax, 0x4c00
    int 0x21

result dw 0
```

CARRY FLAG WILL BE 0 UNTIL AX AND BX ARE ADDED AS THE MSB DOES NOT HAVE A CARRY YET



AFTER THE EXECUTION OF ABOVE COMMAND MSB HAS A CARRY WHICH RAISES THE CARRY FLAG



SIGNED FLAG :

```
1      org 0x0100
2
3      start:
4          mov ax, 00001011b;11
5          mov bx, 00001111b;15
6          sub ax, bx
7          mov [result],ax
8
9          mov ax,0x4c00
10         int 0x21
11
12     result dw 0
```

AX AND BX BOTH STORE POSITIVE BINARY NUMBER TILL THEY ARE SUBSTRACTED

DOSBox 0.74-3,...

Register	Value
AX	0000
SI	0000
CS	19F5
IP	0106
Stack	+0 0000
Flags	7200
DI	0000
DS	19F5
SP	0000
ES	19F5
HS	19F5
FS	19F5

0106:29D8 SUB AX,BX

0106:29D9 MOV [0110],AX

0106:29DA MOV AX,4C00

0106:29DB INT 21

0106:29DC CLD

0106:29DD JMP Far BX

0106:29DE ADD AL,31

0106:29DF DB 02

AS BX (15) IS SUBSTRACTED FROM AX (11) WHICH WILL RESULT INTO A NEGATIVE NUMBER (-4) SIGNED FLAG WILL BE RAISED:

DOSBox 0.74-3,...

Register	Value
AX	FFFF
SI	0000
CS	19F5
IP	0106
Stack	+0 0000
Flags	7295
DI	0000
DS	19F5
SP	0000
ES	19F5
HS	19F5
FS	19F5

0106:29D8 SUB AX,BX

0106:29D9 MOV [0110],AX

0106:29DA MOV AX,4C00

0106:29DB INT 21

0106:29DC CLD

0106:29DD JMP Far BX

0106:29DE ADD AL,31

0106:29DF DB 02

ZERO FLAG:

```

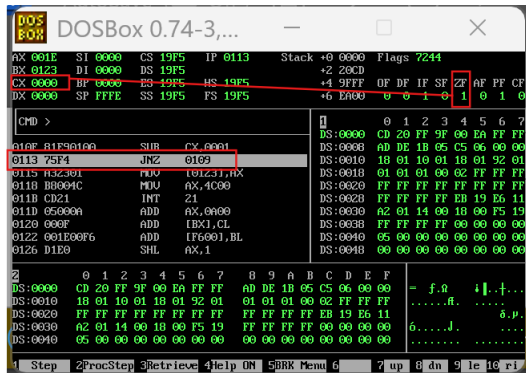
1      [org 0x0100]
2
3      mov     ax, 0
4      mov     cx, 3
5      mov     bx, num1
6      l:
7          add     ax, [bx]
8          add     bx, 2
9
10         sub     cx, 1
11         jnz     l
12
13         mov     [result], ax
14
15         mov     ax, 0x4c00
16         int     0x21
17
18 num1: dw     5, 10, 15
19 result: dw    0

```

ZF IS NOT RAISED YET AS THE LOOP EXECUTION HAVENT MADE CX ZERO

[illegible]

AFTER THE EXECUTION OF PREVIOUS COMMAND CX BECOME 0 AND ZF IS RAISED:



OVERFLOW FLAG:

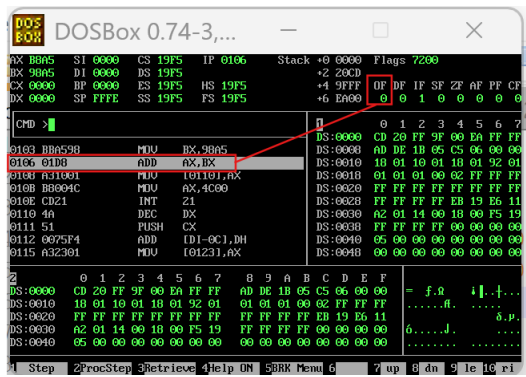
```
org 0x0100

start:
    mov ax, 1011100010100101b
    mov bx, 1001100010100101b
    add ax, bx
    mov [result], ax

    mov ax, 0x4c00
    int 0x21

result dw 0
```

FROM ABOVE EXAMPLE, CF IS NOT RAISED YET AS AX AND BX STORES 16 BIT TILL NOW



A CARRY FROM MSB WOULD CAUSE AN OVERFLOW AS AX CAN STORE ONLY 16 BIT WHICH IS OVERFLOWED AFTER $1+1=10$ FROM MSB

DOSBox 0.74-3,...

AX 514a SI 0000 CS 19F5 IP 0100 Stack +0 0000 Flags 7001
 DX 9005 DI 0000 DS 19F5 +2 20C0
 CX 0000 BP 0000 ES 19F5 HS 19F5 +4 9FFF OF DF IF SF ZF AF PF CF
 DX 0000 SP FFFF SS 19F5 FS 19F5 +6 E000 1 0 1 0 0 0 0 1

CMD >

0100 0100 ADD AX,BX
 0108 A31001 MOV [0110],AX
 010B B0004C MOV AX,4C00
 010E CD21 INT Z1
 0110 40 DEC DX
 0111 51 PUSH CX
 0112 0075F4 ADD [DI-0C],DH
 0115 A32301 MOV [0123],AX
 011B B0004C MOV AX,4C00

DS:0000 CD 20 FF 9F 00 EA FF FF AD DE 1B 05 C5 06 00 00
 DS:0010 1B 01 10 01 1B 01 92 01 01 01 01 00 02 FF FF FF
 DS:0020 FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF
 DS:0030 A2 01 14 00 1B 00 F5 19 FF FF FF FF 00 00 00
 DS:0040 05 00 00 00 00 00 00 00 00 00 00 00 00 00 00

INTERRUPT FLAG:

INTERRUPT FLAG WILL BE 1 UNTIL INT 21 IS EXECUTED ON THE COMPILER

DOSBox 0.74-3,...

AX 4C00 SI 0000 CS 19F5 IP 010E Stack +0 0000 Flags 7001
 DX 9005 DI 0000 DS 19F5 +2 20C0
 CX 0012 BP 0000 ES 19F5 HS 19F5 +4 9FFF OF DF IF SF ZF AF PF CF
 DX 0000 SP FFFF SS 19F5 FS 19F5 +6 E000 1 0 1 0 0 0 0 1

CMD >

0100 B0004C MOV AX,4C00
 010E CD21 INT Z1
 0110 40 DEC DX
 0111 51 PUSH CX
 0112 0075F4 ADD [DI-0C],DH
 0115 A32301 MOV [0123],AX
 011B B0004C MOV AX,4C00
 011D 050000 ADD AX,0000

DS:0000 CD 20 FF 9F 00 EA FF FF AD DE 1B 05 C5 06 00 00
 DS:0010 1B 01 10 01 1B 01 92 01 01 01 01 00 02 FF FF FF
 DS:0020 FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF
 DS:0030 A2 01 14 00 1B 00 F5 19 FF FF FF FF 00 00 00
 DS:0040 05 00 00 00 00 00 00 00 00 00 00 00 00 00 00

AFTER THE EXECUTION INTERRUPT FLAG BECOMES ZERO INDICATING PROGRAM HAS ENDED

DOSBox 0.74-3,...

AX 4C00 SI 0000 CS F000 IP 1400 Stack +0 4200 Flags 7001
 DX 9005 DI 0000 DS 19F5 +2 06C5
 CX 0012 BP 0000 ES 19F5 HS 19F5 +4 7801 OF DF IF SF ZF AF PF CF
 DX 0000 SP FFF2 SS 19F5 FS 19F5 +6 0110 1 0 0 0 0 0 0 1

CMD >

010E CD21 INT Z1
 1400 F0 STI
 1401 FE DB FE
 1402 3025 CMP DI,11,0H
 1404 00C7 ADD BH,CL
 1406 CB RET Far
 1407 51 PUSH CX
 140B B94001 MOV CX,0140
 140D E2FE LOOP 140B

DS:0000 CD 20 FF 9F 00 EA FF FF AD DE 1B 05 C5 06 00 00
 DS:0010 1B 01 10 01 1B 01 92 01 01 01 01 00 02 FF FF FF
 DS:0020 FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF
 DS:0030 A2 01 14 00 1B 00 F5 19 FF FF FF FF 00 00 00
 DS:0040 05 00 00 00 00 00 00 00 00 00 00 00 00 00 00